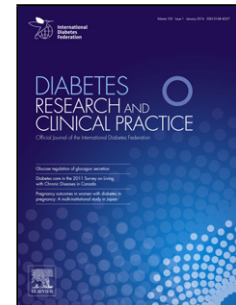


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Interventions to Maintain Cardiac Risk Control After Discharge from a Cardiovascular Risk Reduction Clinic: A Randomized
Controlled Trial

Running title: Interventions to Maintain Cardiac Risk Control

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Abstract

Aims: To evaluate the efficacy of two maintenance strategies compared to usual care after discharge from a pharmacist-led cardiovascular risk reduction clinic (CRRC).

Methods: Open-label, randomized-controlled trial of 200 consecutive CRRC patients that met clinic discharge criteria ($\text{HbA1c} \leq 7\%$ (53mmol/mol); Blood Pressure $\leq 140/80\text{mmHg}$ for those with diabetes and $\leq 140/90\text{mmHg}$ for those without diabetes; and an LDL-cholesterol $\leq 2.59\text{mmol/l}$. Participants were randomized to either (1) quarterly group medical visits or (2) quarterly CRRC individual clinic visits, or (3) a usual care control arm with the standard primary care alone first in a 1:1:1 ratio, followed by a 2:2:1 ratio after first 100 patients. Primary outcome measures were time to failure for guideline recommended goals of HbA1c and blood pressure over 12-months.

Results: Of the 200 participants randomized, 89% had diabetes and were similar in other cardiovascular risk factors. After 1-year, the HbA1c failure rate was 0.36 [95%CI, 0.28-0.47] per quarter for the group medical visit arm, 0.24 [95%CI, 0.18-0.33] per quarter for the quarterly CRRC individual arm and, 0.82 [95%CI, 0.69-0.96] per quarter for the usual care control arm, $p<0.001$. The rate of failure for blood pressure was 0.31 [95%CI, 0.23-0.41] per quarter for the group medical visit arm, 0.22 [95% I, 0.16-0.30] per quarter for the CRRC individual arm and, 0.53 [95%CI, 0.40-0.71] per quarter the control arm, $p<0.001$.

Conclusion: After discharge from a CRRC program, both individual and group interventions are more effective in maintaining glycemia and blood pressure control for patients with diabetes than usual care after 1-year of follow-up.

Diabetes mellitus is a chronic and progressive illness that affects nearly 26 million Americans.(1, 2) Several clinical trials have underscored the importance of attaining and maintaining glycemic control along with control of other major modifiable cardiovascular disease (CVD) risk factors in order to reduce micro and macrovascular co-morbidities.(3-7) However, much of the available data describes care delivery models for attainment of risk factor control and little data is available as how to best coordinate care interventions to prevent the loss of glycemic and CVD risk factor control once it is achieved. (8-10)

The Cardiovascular Risk Reduction Clinic (CRRC) is a pharmacist-coordinated care model comprised of monthly individual clinic visits with a pharmacist in addition to their standard primary care that integrates the management of the hypertension, hyperlipidemia and hyperglycemia into a single treatment plan to achieve the therapeutic goals recommended by the national guidelines.(8-10) The effectiveness of the CRRC in attaining guideline recommendations for treatment of CVD risk factors has been previously described and established.(11) Since control of CVD risk factors from these type of specialized intensive care models are known to relapse over time;(12) there is a need to identify strategies to maintain target goals for CVD risk factors after discharge from an intensive care intervention program such as CRRC. Group medical visits have been proposed as an alternative mode of health care delivery and have demonstrated efficacy in attainment of target levels for glycemia and CVD risk factors.(13-15) However, little data is available comparing the efficacy of this alternative model against traditional health care delivery interventions for the maintenance of risk factor control once it is attained.

The objective of this study is to evaluate the efficacy of two cardiovascular risk maintenance interventions, 1) continued quarterly follow-up individual visits in CRRC, 2) quarterly group medical visits as compared to 3) usual care (no further intensive program), after discharge from the CRRC at the Providence VA Medical Center (VAMC).

Research Design, Methods and Patients

Study Design

We performed an open-label, randomized controlled trial of 200 consecutive CRRC patients that met clinic discharge criteria (HbA1c $\leq 7\%$ (53mmol/mol); Blood Pressure $\leq 140/80$ mmHg for those with diabetes and $\leq 140/90$ mmHg for those without diabetes; and an LDL cholesterol ≤ 2.59 mmol/l obtained within 30 days of their final CRRC visit).(8-10) This trial was registered with Clinicaltrials.gov public database, a registry and results database of publicly and privately supported clinical studies of human participants supported by the U.S. National Institutes of Health (ClinicalTrials.gov Identifier: NCT00358033). Written informed consent was obtained from each participant prior to randomization. Participants were randomized to either (1) quarterly group medical visits or (2) quarterly CRRC individual clinic visits (as opposed to monthly during the intensive program) in addition to standard primary care, or (3) a control arm with the standard primary care alone, initially in a 1:1:1 ratio. However, after the first 100 participants were enrolled, it became evident that the intervention arms would perform better than the control arm. With the permission of the Providence VA Medical Center Institutional Review Board, we changed our randomization strategy to a 2:2:1 ratio so that we could over sample the two intervention arms in order to detect potential differences between the two interventions

strategies. The Institutional Review Board and Research and Development Committees at the Providence VAMC approved the protocol. All study procedures were conducted in accordance with the ethical standards of the Helsinki Declaration of 1975. Enrollment for this study began October 2003 and ended December 2006 (Figure 1). This trial was sponsored by the Merck and Co. Inc. Disease State Management Grant Program in conjunction with the Providence VA Medical Center.

Study population.

Patients were eligible if they were actively enrolled in the CRRC clinic, had documented CVD or diabetes mellitus and achieved the aforementioned CRRC clinic discharge criteria. Patients were excluded if they had a condition that may limit long term adherence to study visits or life-expectancy less than one year, such as severe dementia, acute psychiatric decompensation within the previous 6 months, unstable psychiatric illness, metastatic cancer or a terminal illness.

Interventions

Group Medical Visits: In addition to attending regular visits with a primary care provider, participants randomized to group medical visits attended a total of four 120 minute sessions held every 3 months over a period of 12 months. Each session included approximately 6-8 participants. Participants were encouraged to bring their family members or members of social support to these sessions. These sessions were facilitated by a clinical pharmacist and consisted of education during the first hour of the intervention, followed by behavioral and pharmacological interventions during the second hour for hyperglycemia, hypertension and dyslipidemia. The educational component included interactive lectures that were based on the American Diabetes Association (ADA) Standards of

Diabetes Self-Management and focused on one or two of the core diabetes content areas such as health eating or physical activity as identified by the 2006 National Diabetes Self-Management Education Task Force. (9, 16) The clinical pharmacists who delivered these group interventions were diabetes core content experts and certified as diabetes educators.

During the second hour of the group medical visits, each participant was provided with an individualized cardiovascular risk report card that contained their medical history, current medications, vitals and laboratory values obtained prior to the sessions. These report cards were reviewed with participants in the group visit session and updated every 3 months with laboratory measurements that were drawn throughout the study such as a repeat lipid panel or an HbA1c. Based on these report cards, medications for blood pressure, cholesterol and diabetes were initiated or titrated in the group visit session in accordance with previously established treatment algorithms that were consistent with the VHA Hospitals' national formulary.(17) Each group member was provided with individualized homework for medication changes, and a behavior change goal, such as exercise recommendations, dietary modifications and blood glucose or blood pressure monitoring. The number of behavioral change goals was negotiated with each individual in the group session and based on the participants' need and their ability to perform these self-care behaviors. Demonstration and coaching were used to increase the frequency of self-care skills such as monitoring of blood glucose and logging daily dietary intake.(18) Individual breakout sessions between the pharmacist and group participants were not performed; however, participants were contacted via telephone on an as needed basis to follow-up on pertinent laboratory values or to reinforce self-care monitoring or medication changes.

CRRC Individual Clinic Sessions: The standard CRRC sessions consisted of 30 minute visits with a clinical pharmacist held once every 2-6 weeks until cardiovascular risk control has been attained. At each of the clinic visits, the clinical pharmacists assessed medication adherence, obtained vitals and laboratory parameters and titrate medications to address blood pressure, hyperlipidemia and diabetes. Participants were referred to a nutritionist or physical therapist for an individualized diet and exercise programs on an as needed basis. Patients are discharged from CRRC, back to usual primary care, once they attained the cardiovascular risk factor goals. For this study, if the patients are randomized to the CRRC arm after cardiovascular risk goal attainment, they will be followed every 3 months, instead of the usual 2-6 weeks, over the one year study period for a total of 4 sessions.

Standard Primary Care Follow-up Control: Participants continued with their standard care with a primary care provider which occurred on an average of 3-4 times a year for patients with chronic diseases such as diabetes. The standard of care with primary care providers at the Providence VAMC was delivered through individual contact. Laboratory parameters and vital signs were obtained at the discretion and frequency of the participant's own primary care provider. The primary care providers had referral access to nutrition and physical therapy and all of the same consultation services as the CRRC clinic provider. However, referrals back to the CRRC clinic were not accepted during the one year study follow-up.

Measurements and Outcomes

Since current treatment guidelines differ for those with diabetes and without diabetes despite presence of CVD, the outcomes for these patients were presented separately.(8, 10, 19)

For patients with diabetes, our primary outcome measures were time to failure for guideline recommended goals of HbA1c and blood pressure over the 12 month study period. Glycemic control and blood pressure were chosen as our primary outcome measures since studies have demonstrated that there is a progressive increase in the risk of both macrovascular and microvascular events with increasing levels of HbA1c and blood pressure control.(20, 21) Consistent with the American Diabetes Association Standards of Medical Care recommendations, failure to maintain guideline goals was defined as an HbA1c $>7\%$ ($>53\text{mmol/mol}$) or a blood pressure value $>140/80\text{mmHg}$.(19) All HbA1c tests were performed at Providence VAMC clinical laboratory utilizing standard high-pressure liquid chromatography methods with a coefficient of variation (CV) of 1.1% within control data (accurate results are considered to be $<3\%$ CV); therefore we expected the true HbA1c value to lay within a standard deviation of $\pm 0.1\%$ for an HbA1c between 5.0-10.0% (31-86 mmol/mol). Blood pressure was measured in the usual clinic settings by clinical staff, with the participant in the seated position using an automatic electronic cuff. (20). For patients without diabetes, failure to maintain guideline recommended goals was defined as a blood pressure $>140/90\text{ mmHg}$.(10)

Secondary outcome measures were the maintenance of guideline recommended goals for LDL cholesterol at 12 months for both patients with and without diabetes. Guideline goals for LDL cholesterol was defined as obtaining at least one fasting LDL cholesterol level of $\leq 2.59\text{mmol/l}$ over the one year follow-up.(21) LDL cholesterol was calculated using the Friedewald equation.(22) The coefficients of variation for total cholesterol, HDL cholesterol, and triglycerides were 1.80%, 1.23%, and 3.93%, respectively.(23)

Process outcomes: We also collected the change in the number of prescribed medications for hypertension, hyperlipidemia or hyperglycemia from baseline and visits to primary care, and specialty care. Adverse event rates during the study period were tracked: hospital admissions and visits to the emergency department for each study arm.

Data Collection: Demographic variables, co-morbid health conditions, tobacco use and active medication use were obtained from the electronic medical record at baseline and at the end of the study. The total number of primary care, diabetes specialty care (i.e. cardiology and endocrinology), diabetes preventive care (i.e. podiatry, ophthalmology/optometry, diabetes self-management education, nutrition, exercise therapy, and tobacco cessation) and group medical visits or individual CRRC visits were tracked for all study participants. Emergency department (ED) care, hospitalizations, and death were defined as serious adverse events and were also tracked for all study participants.

Statistical Analysis

Cohort of Patients with Diabetes

Baseline characteristics amongst the three study arms were compared using ANOVA for continuous variables and chi-square for discrete variables. The failure rates quantified by quarters for HbA1c and blood pressure goals were plotted in Kaplan-Meier curves and compared using log rank tests amongst the three study arms. Jackknife method to estimate variance was used when the study sample falls <5 subjects during follow-up. Cox proportional-hazards models were used to adjust for study characteristics which were imbalanced at baseline in the comparison of failure rates of HbA1c and blood pressure goals amongst the three study arms using the

standard care arm as the reference. We tested the proportionality of hazards assumption for the Kaplan-Meier curves by analysis of the Schoenfeld Residuals.(24) The proportional hazards assumptions was confirmed for HbA1c ($p=0.62$) and blood pressure ($p=0.97$) analyses.

To analyze maintenance of LDL cholesterol goals at 12 months, we used logistic regression modeling to adjust for imbalances in baseline characteristics.

Analysis of process outcomes: the number of hospital admissions, emergency department, primary care and specialty care visits over the study period were compared using linear regression and adjusted for imbalanced baseline characteristics. The change in the number of total prescribed medications for hypertension, hyperlipidemia or hyperglycemia from baseline was compared using a paired t-test for each of the study arms.

Cohort of Patients without Diabetes

Given small sample of patients without diabetes (please see below in results), the maintenance of blood pressure and LDL cholesterol goals for those participants with CVD but without diabetes were presented descriptively.

All analyses were done on an intention-to-treat basis and performed using Stata SE 11.0 (StataCorp, College Station, TX).

Results

Participants and Attendance

Of the 740 patients seen in the CRRC clinic between October 2003 and December 2006, 540 patients met eligibility criteria, 200 patients (37%) agreed to participate and were randomly assigned. (Figure 1) Of the 200 study participants who were randomized, 89% (n=178) of them had diabetes and 11% (n=22) had CVD but no diabetes. Complete one year follow-up data was available on 181 of participants (90.5%).

Participants with Diabetes

The characteristics of the study population with diabetes are shown in Table 1. There was a trend for CRRC individual and group medical visit participants to be a current smoker (CRRC individual, 17.2% vs. group, 18.0% vs. standard care, 7.6%, $p=0.19$) and diagnosed with a co-morbid mental health condition (CRRC individual, 50.0% vs. group, 54.1% vs. standard care, 34.0%, $p=0.08$) at baseline; but were otherwise similar in other cardiovascular risk factor values.

Primary Outcomes

Hemoglobin A1c failure rates: Over the one year follow-up period, the rate of failure from the HbA1c goal was 0.24 [95% CI, 0.18 to 0.33] per quarter out of 156 person-quarters of follow-up for the CRRC individual arm, 0.36 [95% CI, 0.28 to 0.47] per quarter out of 129 person-quarters of follow-up for the group medical visit arm and, 0.82 [95% CI, 0.69 to 0.96] per quarter out of 60 person-quarters of follow-up for the standard care arm, $p<0.001$. Post hoc analysis comparing the failure rates between two intervention arms using log-rank tests showed the CRRC individual arm to be similar to the group intervention arm ($p=0.07$). After adjustment for smoking and co-morbid mental health conditions, multivariate Cox models showed statistically significant reduction in failure rates

for participants in the CRRC individual (HR, 0.34 [95% CI 0.21 to 0.53]) and the group medical visit intervention arms (HR, 0.49 [95% CI 0.32 to 0.75]) vs. standard care, $p < 0.001$. (Figure 2)

Blood pressure failure rates: The rate of failure for blood pressure goals was 0.22 [95% CI, 0.16 to 0.30] per quarter out of 166 person-quarters of follow-up for the CRRC arm, 0.31 [95% CI, 0.23 to 0.41] per quarter out of 140 person-quarters of follow-up for the group medical visit arm and, 0.53 [95% CI, 0.40 to 0.71] per quarter out of 87 person-quarters of follow-up for the standard care arm, $p < 0.001$. Post hoc analysis comparing the failure rates between two intervention arms using log-rank tests showed the CRRC arm to be similar to the group intervention arm ($p = 0.10$). After adjustment for smoking and co-morbid mental health condition, Cox modeling showed statistically significant reductions in failure rates for the CRRC (HR, 0.43 [95% CI, 0.27 to 0.68]) and group intervention (HR, 0.62 [95% CI, 0.41 to 0.95] arms vs. the standard care arm, $p < 0.002$. (Figure 2)

Secondary Outcomes

The proportion of participants that were guideline adherent for LDL cholesterol after one year of follow-up was not different for the three treatment arms (75.0% for CRRC individual, vs. 88.5% for group, vs. 84.3% for standard care, $p = 0.12$). After adjustment for baseline differences in smoking and mental health conditions, logistic regression modeling confirmed these findings.

Only 1 participant in the CRRC arm reported tobacco cessation after 1 year of follow-up; whereas no participants quit smoking in the group visit intervention or standard care arms.

Process Measures

Over the 12 month study period, participants in the CRRC and group visit interventions had a similar number of primary care providers visits (CRRC 2.8 ± 1.1 visits, group intervention 3.0 ± 1.2 visits vs. standard care, 2.8 ± 1.0 visits, $p=0.46$). Participants in the CRRC arm were more likely to attend study visits than in the group intervention arm (CRRC 3.4 ± 0.9 visits vs. group intervention 2.8 ± 1.4 visits, $p=0.010$) over the 1 year study period. Participants in the CRRC individual and group visit arms were more likely to attend diabetes preventive care visits (ophthalmology, podiatry, nutrition, tobacco cessation, and physical therapy exercise) as compared to standard care participants (CRRC individual 3.8 ± 2.4 visits; group 4.1 ± 2.6 visits vs. standard care 2.7 ± 2.1 visits, $p=0.01$) over the 12 month study period.

Medication Use

Overall, the total number of prescribed medications for hypertension, hyperlipidemia or hyperglycemia did not change significantly from baseline for CRRC participants (Table 2). However, the total number of prescribed medications for hypertension and hyperglycemia significantly increased during the study period for group medical visit participants. The number of active prescriptions for hyperlipidemia and hyperglycemia decreased significantly from baseline for standard care participants.

Adverse Events

The average number of emergency department visits were comparable among the three treatment arms (CRRC individual 0.4 ± 0.8 visits; group 0.6 ± 1.0 visits vs. standard care, 0.6 ± 1.1 visits, $p=0.35$). Similarly, the average number of hospital admissions (CRRC individual, 0.3 ± 0.7 admissions; group 0.4 ± 0.8 admissions vs. standard care 0.2 ± 0.5 admissions, $p=0.18$) were also

comparable for all three study arms over the course of the study period. Adjustment for imbalances in baseline smoking status and diagnosis of a co-morbid mental health condition did not alter these findings.

Cohort of Patients without Diabetes

Participants with CVD alone (n=22) were younger (mean of 62.7 ± 8.9 years) but had a similar level of co-morbid conditions and baseline risk factor control (blood pressure $121.4 \pm 11.6/70.0 \pm 9.3$ mmHg and LDL cholesterol 2.09 ± 0.39 mmol/l) as compared to the overall study population (Supplemental Table S1). Of those patients with CVD alone, 9 were randomized to CRRC intervention arm, 11 were randomized to the group intervention arms and 2 were randomized to standard care arm. After one year of follow-up, 66.7% (n=6) of participants in the CRRC arm, 54.6% (n=6) of participants in the group intervention arm and 100% (n=2) of participants maintained a blood pressure $<140/90$ mmHg. Maintenance of LDL cholesterol ≤ 2.59 mmol/l was achieved by 44.4% (n=4) of participants in the CRRC arm, 36.4% (n=4) of participants in the group intervention arm and 50% (n=1) of participants with CVD alone.

Conclusions

After goal attainment and discharge from an intensive cardiovascular risk reduction program, both pharmacist-led CRRC individual clinic visit and group visit interventions are more effective than usual care in maintaining glycemic and blood pressure control over a year for patients with diabetes. Between the two active interventions, CRRC individual clinic visits have similar failure

rates for guideline recommended glycemic and blood pressure control as compared to group visit interventions. Thus, for patients with diabetes, less intensive, quarterly follow-up visits, delivered as individual or group visits, may be considered as efficacious alternative health care models to maintain glycemic and CVD risk factor control after discharge from an intensive treatment program. However, these differences did not translate into a significant reduction in emergency room utilization or hospitalizations as compared to the standard care arm over the one year of follow-up.

These findings are important because the capacity of the current health system to provide chronic disease care for an aging population with an expanding diabetes epidemic is diminishing, as the number of adult primary care providers declines and the demand for provider visits increases.(25, 26) Further, our previous data demonstrated the need to identify optimal maintenance strategies after risk factor control is achieved since more than 50% fail to maintain systolic blood pressure control 3 months after and 25% fail to maintain glycemic control 9 months after discharge from an intensive treatment program.(29) However, scant research is available to support an optimal strategy for glycemic and CVD risk factor maintenance and speak to the novelty of our findings, since little data exist comparing the use of a group intervention strategy versus traditional care strategies for the maintenance of risk factor control.(30) Based on our findings, more than half of the primary care provider's time allotted for chronic disease management such as patient education, counseling on lifestyle changes and medication adherence, could be undertaken in a less costly manner by non-physician providers in these alternative care settings. Our results also showed that a return to a less intensive model of care without additional aid will result in a faster decline of diabetes and cardiovascular risk control. We have shown in the past that pharmacist-led cardiovascular risk reduction clinic (CRRC) was efficacious in improving diabetes and cardiovascular risk factors.(11, 27) Therefore,

the efficacy of a similar pharmacist-led individual clinic model to maintain diabetes and cardiovascular risk goals in a less intensive program, is albeit novel, but not surprising. In contrast, the efficacy data for group medical visits to treat diabetes and cardiovascular risk factors is more controversial,⁽²⁸⁾ and the efficacy of this group treatment model to maintain risk factor control after discharge from an intensive individual clinic has been uncertain.

Group medical visits have been proposed as an alternative mode of health care delivery to improve key elements of care, in particular access and outcomes.^(29, 30) Improved access is thought to occur simply because the amount of time it takes to simultaneously see all patients in the group setting is usually shorter than the time it would take to see each patient in the individual setting, thereby improving the provider's productivity and access. Improved outcomes may occur in the group setting as compared to standard care because (1) the group setting may provide an enhanced diabetes and CVD self-management training due to the peer support of group members toward lifestyle changes when a group of patients has similar disease management goals, (2) the group also provides a focused environment for care of the common condition or unifying characteristic (e.g., diabetes complications) without the competing demands of multiple other issues that come up in brief primary care visits, and/or (3) the group provides access to medication changes performed by a provider with special expertise in diabetes and CVD risk reduction, thus the group may serve as a specialty referral.⁽²⁸⁾ Furthermore, costs are thought to be lowered because the same education and training are given to a group as opposed to only one individual thereby improves efficiency and access. Additional lower costs of care can be achieved by preventing patients from using acute-care or emergency department settings for management of their chronic illness.⁽³¹⁾ However, differences in the interim acute care or emergency department visits were not appreciated in our study over the one year of follow-up. Cost may

also be lowered in our proposed group medical visit model through the use of alternative non-physician providers, in this case, clinical pharmacists, who can perform both disease state management education in addition to medication changes in lieu of physicians in a one-stop fashion; although we did not formally track costs in our study.

Our study has limitations. First, the generalizability of our study may be limited since it was conducted at a single site and included only veterans, a cohort that routinely receives care from non-physician providers and care interventions in a group setting.(32, 33) Nevertheless, group medical visits have been shown to be efficacious in HbA1c goal attainment in the non-veteran populations.(34) But, repeating this study in a non-veteran population would be important to understand the feasibility of these interventions in populations that do not historically receive care in these alternative settings. Second, costs were not formally tracked. Future work will need to involve formal cost analyses in order to determine how these alternative care delivery models may be cost saving while maintaining cardiovascular risk goals. Last, we are underpowered to make any conclusions on patients with CVD but without diabetes who were enrolled in the study, a population that too deserve future studies to maintain risk control once these are achieved.

In closing, both pharmacist-led individual and group medical visits for patients with diabetes offer the promise of improving the effectiveness and possibly efficiency (for group visits) of health care. The results presented suggest that implementation of the alternative care models to prevent loss of glycemic and CVD risk factor control in patients with diabetes is feasible and efficacious.

Author contributions

Both Drs. Taveira and Wu oversaw study conduction, analyzed the data, contributed to the analysis, prepared and edited the manuscript.

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Conflict of interest

The authors declare that they have no conflict of interest.

Disclaimer

The views expressed in this article are those of the authors and do not necessarily reflect the position or policy of the Department of Veterans Affairs or the United States government.

References

1. UKPDS Study Group. Intensive blood-glucose control with sulphonylureas or insulin compared with conventional treatment and risk of complications in patients with type 2 diabetes (UKPDS 33). *Lancet*. 1998;352(9131):837-53.
2. Centers for Disease Control. National diabetes fact sheet: national estimates and general information on diabetes and prediabetes in the United States. In: Centers for Disease Control, editor. Centers for Disease Control. Atlanta, GA 2011.
3. Roger VL, Go AS, Lloyd-Jones DM, Adams RJ, Berry JD, Brown TM, et al. Heart Disease and Stroke Statistics-2011 Update: A Report From the American Heart Association. *Circulation*. 2011;123(4):e18-209. doi: 10.1161/CIR.0b013e3182009701.
4. Gaede P, Vedel P, Larsen N, Jensen GVH, Parving H-H, Pedersen O. Multifactorial Intervention and Cardiovascular Disease in Patients with Type 2 Diabetes. *New England Journal of Medicine*. 2003;348(5):383-93.
5. Holman RR, Paul SK, Bethel MA, Matthews DR, Neil HA. 10-year follow-up of intensive glucose control in type 2 diabetes. *N Engl J Med*. 2008;359(15):1577-89.
6. Ray KK, Seshasai SRK, Wijesuriya S, Sivakumaran R, Nethercott S, Preiss D, et al. Effect of intensive control of glucose on cardiovascular outcomes and death in patients with diabetes mellitus: a meta-analysis of randomised controlled trials. *The Lancet*. 2009;373(9677):1765-72.

7. Wu Hong, Xu Mao-jin, Zou Da-jin, Han Qiao-jun, Xue H. Intensive glycemic control and macrovascular events in type 2 diabetes mellitus: a meta-analysis of randomized controlled trials
Chinese Medical Journal. 2010;123(20):2908-13.
8. Expert Panel on Detection Evaluation and Treatment of High Blood Cholesterol in Adults. Executive Summary of the Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III). JAMA. 2001;285(19):2486-97.
9. American Diabetes Association. Standards of Medical Care in Diabetes-2006. Diabetes Care. 2006;29(S1):S4-42.
10. Chobanian A, Bakris G, Cushman W, Green L, al. e. Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III) NHLBI. Hypertension. 2003(42):1206-52.
11. Taveira TH, Wu WC, Martin OJ, Schleinitz M, Friedman P, Sharma S. Pharmacist-led cardiac risk reduction model. Preventive Cardiology. 2006;9(4):202-8.
12. Pirraglia P, Taveira TH, Cohen L, Wu W. Effectiveness of a multifactorial cardiovascular risk reduction clinic for diabetes patients with depression Preventing Chronic Disease. 2008;5(4):1-7.
13. Taveira TH, Friedmann PD, Cohen LB, Dooley AG, Khatana SAM, Pirraglia PA, et al. Pharmacist-Led Group Medical Appointment Model in Type 2 Diabetes. The Diabetes Educator. 2010;36(1):109-17.

14. Cohen LB, **Taveira TH**, Khatana SA, Dooley AG, Pirraglia PA, Wu WC. Pharmacist-led shared medical appointments for multiple cardiovascular risk reduction in patients with type 2 diabetes. *Diabetes Educator* 2011 Nov;37(6):801-12.
15. Edelman D, Fredrickson SK, Melnyk SD, Coffman CJ, Jeffreys AS, Datta S, et al. Medical Clinics Versus Usual Care for Patients With Both Diabetes and Hypertension. *Annals of Internal Medicine*. 2010;152(11):689-96.
16. Mensing C, Boucher J, Cypress M, Weinger K, Mulcahy K, Barta P, et al. National Standards for Diabetes Self-Management Education. *Diabetes Care*. 2006;29(90001):S78-S85.
17. Willett WC, Howe GR, Kushi LH. Adjustment for total energy intake in epidemiologic studies. *Am J Clin Nutr*. 1997;65(4 Suppl):1220S-8S; discussion 9S-31S.
18. Rollnick S, Mason P, Butler C. Health behavior change: a guide for practitioners. New York: Churchill Livingstone; 1999.
19. American Diabetes Association. Standards of Medical Care in Diabetes 2013. *Diabetes Care*. 2013;36(Supplement 1):S11-S66.
20. Tight blood pressure control and risk of macrovascular and microvascular complications in type 2 diabetes: UKPDS 38. UK Prospective Diabetes Study Group. *BMJ* 1998;317:703-13.
21. Stratton IM, Adler AI, Neil HA, Matthews DR, Manley SE, Cull CA, Hadden D, Turner RC, Holman RR. Association of glycaemia with macrovascular and microvascular complications of type 2 diabetes (UKPDS 35): prospective observational study. *BMJ* 2000;321:405-12.

22. Kim JW, Bosworth HB, Voils CI, Olsen M, Dudley T, Gribbin M, et al. How well do clinic-based blood pressure measurements agree with the mercury standard? *J Gen Intern Med*. 2005;20(7):647-9.
23. National Cholesterol Education Program. Third Report of the Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III). NHLBI 2004 [cited 2005 July 6]. Available from: http://hin.nhlbi.nih.gov/cvd_frameset.htm.
24. Friedewald WT, Levy RI, Fredrickson DS. Estimation of the concentration of low-density lipoprotein cholesterol in plasma, without use of the preparative ultracentrifuge. *Clin Chem*. 1972;18(6):499-502..
25. Bittner V, Olson M, Kelsey SF, Rogers WJ, Bairey Merz CN, Armstrong K, et al. Effect of coronary angiography on use of lipid-lowering agents in women: a report from the Women's Ischemia Syndrome Evaluation (WISE) study. For the WISE Investigators. *Am J Cardiol*. 2000;85(9):1083-8.
26. Schoenfeld D. Partial residuals for the proportional hazards regression model. *Biometrika*. 1982;69:239-41.
27. Bodenheimer T, Pham H. Primary care: current problems and proposed solutions. *Health Aff (Millwood)*. 2010;29:799-805.
28. Massachusetts Medical Society. Access to health care in Massachusetts: the implication of health care reform. 2011.
29. Cohen, LB, Taveira, TH, Wu, WC, Pirraglia, PA. Maintenance of risk factor control in diabetic patients with and without mental health conditions after discharge from a cardiovascular risk reduction clinic. *Ann Pharmacotherapy*. 2010; 44(7-8): 1164-1170.
30. Edelman DA, United States. Department of Veterans Affairs. Health Services Research and Development Service., Durham VA Medical Center. Evidence-based Synthesis Program Center. Shared medical appointments for chronic medical conditions a

systematic review. Washington, DC: Dept. of Veterans Affairs; 2012. Available from:

<http://www.ncbi.nlm.nih.gov/books/NBK99785/>.

31. Taveira TH, Pirraglia P, Cohen L, WC W. Efficacy of a pharmacist-led cardiovascular risk reduction clinic for diabetic patients with and without mental health conditions. *Preventive Cardiology*. 2007;11(4):195-200.
32. Noffsinger E. Increasing quality care and access while reducing cost through drop-in group medical appointments (DIGMAs). *Group Practice Journal*. 1999;48(1):12-8.
33. Housden L, Wong ST, Dawes M. Effectiveness of group medical visits for improving diabetes care: a systematic review and meta-analysis. *CMAJ*. 2013;185(13).
34. Riley SB, Marshall ES. Group visits in diabetes care: a systematic review. *Diabetes Educator*. 2010;36(6):936-44.
35. Cohen LB, Taveira TH, Khatana SA, Dooley AG, Pirraglia PA, WC. W. Pharmacist-led Shared Medical Appointments for Multiple Cardiovascular Risk Reduction in Patients with Type 2 Diabetes. *Diabetes Educator*. 2011;37(6):801-12
36. Taveira TH, Cohen LB, Dooley AG, Khatana SA, WC WC. Pharmacist-led group medical appointments for the management of type 2 diabetes with co-morbid depression in older adults. *Ann Pharmacother* 2011 Nov;45(11):1346-55.
37. Sadur CN, Moline N, Costa M, Michalik D, Mendlowitz D, Roller S, et al. Diabetes management in a health maintenance organization. Efficacy of care management using cluster visits. *Diabetes Care*. 1999;22(12):2011-7.

Table 1. Baseline Characteristics of the Cohort with Diabetes

	CRRC Individual Clinic Sessions	Group Medical Visits	Standard Care	p-value vs. Standard Care
n	64	61	53	
Age (years)	64.6±10.0	64.5±10.2	66.6±10.2	0.29
Female (%)	3.1	1.6	3.8	0.76
African American (%)	7.8	4.9	1.9	0.32
Hypertension (%)	84.1	83.6	90.6	0.48
Type 2 Diabetes Mellitus (%)	95.3	98.4	98.1	0.54
Type 1 Diabetes Mellitus (%)	4.7	1.6	1.9	0.54
Prior MI (%)	15.6	19.7	11.3	0.47
Prior Stroke (%)	6.4	4.9	1.9	0.46
Current Smoker (%)	17.2	16.4	7.6	0.27
BMI Kg/M ²	32.2±6.0	32.6±5.7	31.8±5.5	0.74
SBP mmHg	123.0 ±10.0	121.3 ±8.1	121.0 ±10.2	0.48

DBP mmHg	68.3 ±8.1	68.4 ±9.5	69.3 ±7.8	0.81
Blood Pressure <140/80 mmHg	96.9	96.7	96.2	0.98
HbA1c %	6.3 ± 0.7	6.4 ± 0.5	6.4 ± 0.5	0.39
HbA1c ≤ 7% (53mmol/mol)	93.8	95.1	94.3	0.95
Total cholesterol mmol/l	3.92 ±0.58	4.01 ±0.78	3.83 ±0.59	0.38
HDL cholesterol mmol/l	1.10 ± 0.35	1.04 ± 0.27	1.00 ± 0.23	0.17
Triglycerides mmol/l	1.82 ±1.16	1.87 ± 0.92	2.40 ± 4.49	0.42
LDL cholesterol mmol/l	1.99 ±0.47	2.09 ±0.50	2.04 ±0.42	0.48
LDL ≤ 2.59mmol/l (%)	89.1	91.8	96.2	0.32
Co-morbid Mental Health Condition (%)	50.0	54.1	34.0	0.08
Bipolar Disorder, Schizophrenia or Schizoaffective Disorder (%)	7.8	8.2	7.6	1.00

Values expressed as mean ± standard deviation unless otherwise noted

Table 2. Comparison of Percent of Prescribed Medications at Baseline and Study End for Cohort with Diabetes Mellitus

	Baseline			Study End		
	CRRC Individual Clinic Sessions	Group Medical Visits	Standard Care	CRRC Individual Clinic Sessions	Group Medical Visits	Standard Care
N	64	61	53	64	61	53
Aspirin	79.7	68.3	73.6	84.4	75.4	84.9 [†]
ACE-Inhibitor	68.8	70.5	69.8	72.1	67.9	62.7
Angiotensin II Receptor Blocker	23.8	11.5	30.2	27.9	19.6 [†]	37.2 [†]
Beta-Blocker	56.3	60.7	56.6	59.0	67.9	58.8
Diuretic	51.6	44.3	50.9	57.4	42.9	56.0
Calcium Channel Blocker	33.3	23.0	32.1	24.6 [†]	35.7 [†]	31.3
Sulfonylurea	50.0	43.3	66.0	45.9	42.9	56.9 [†]
Metformin	59.4	53.3	56.6	60.7	58.9	55.8
Thiazolidinedione	35.9	32.8	37.7	36.1	39.3	35.3
Insulin	40.6	36.1	26.4	42.6	41.1	29.4
Statin	79.7	85.2	92.5	82.0	83.9	92.2
Ezetimibe	1.6	1.6	1.9	1.6	1.8	0
Niacin	1.6	9.8	13.2	3.3	14.3	11.5
Fibrate/BAS/Fish Oil	10.9	19.7	20.8	25.0 [†]	23.2	13.7 [†]
Total Cholesterol Medications (mean±SD)	0.9 ± 0.6	1.2 ± 0.6	1.3 ± 0.6	1.1 ± 0.6	1.2 ± 0.6	1.2 ± 0.5 [†]
Total Antihypertensive Medications (mean±SD)	2.3 ± 1.2	2.1 ± 1.2	2.4 ± 1.0	2.4 ± 1.2	2.3 ± 1.3 [†]	2.5 ± 1.2
Total Antihyperglycemic Medications (mean±SD)	1.9 ± 1.0	1.7 ± 1.0	1.9 ± 0.8	1.9 ± 1.0	1.8 ± 1.0 [†]	1.8 ± 0.8 [†]
Values expressed as percent unless otherwise noted; † p<0.05 vs. baseline						

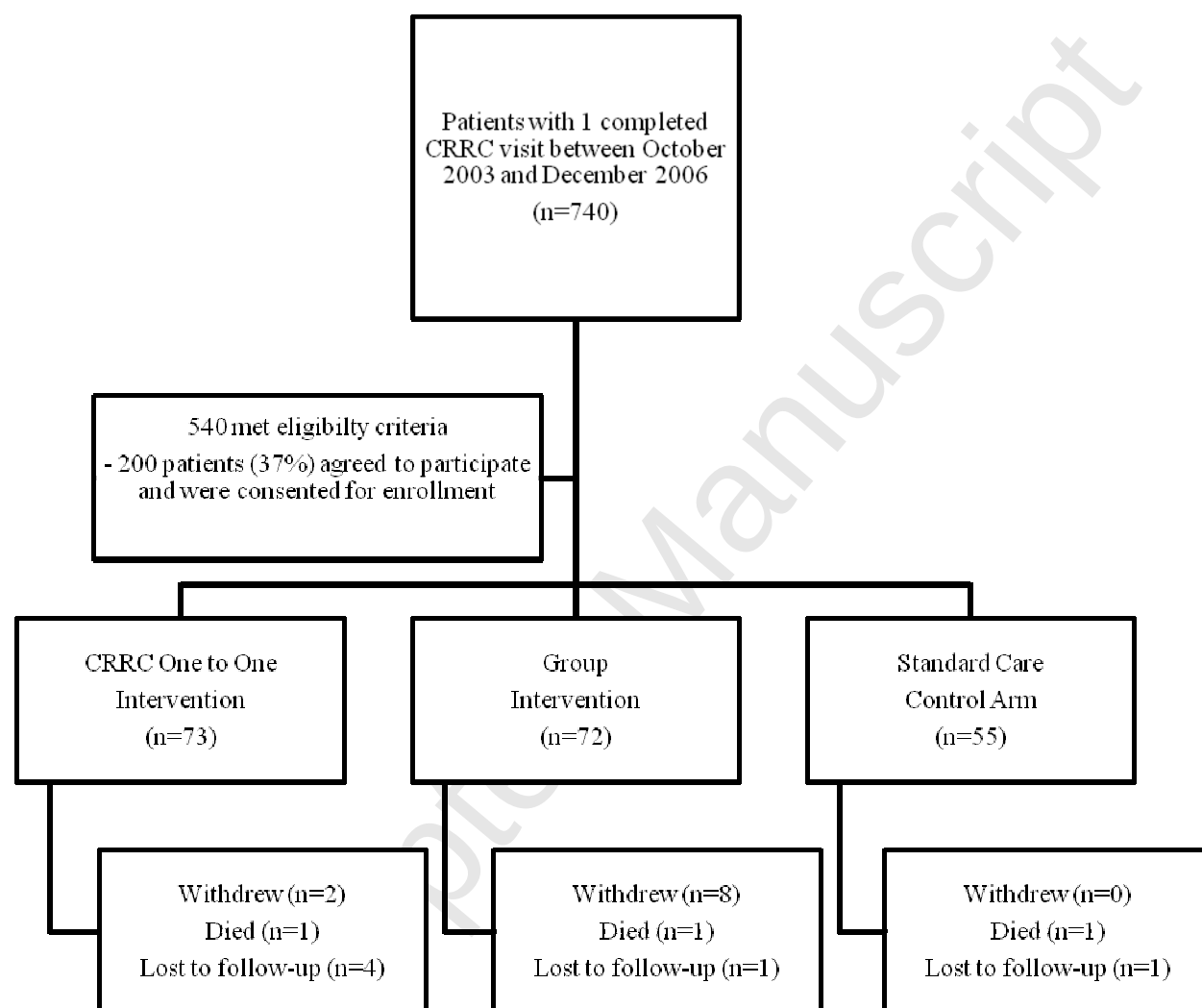
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Figure Legends

Figure 1: Flow Chart of Participants

Figure 2: Kaplan Meier Survival Curves for HbA1c and Blood Pressure Guideline Adherence for Participants with Diabetes Mellitus. Failure to maintain HbA1c guideline adherence was defined as an HbA1c $>7\%$ (53mmol/mol). Failure to maintain blood pressure guideline adherence for participants with Diabetes Mellitus was defined as a blood pressure value $>140/80\text{mmHg}$.

1 Figure 1: Flow Chart of Participants

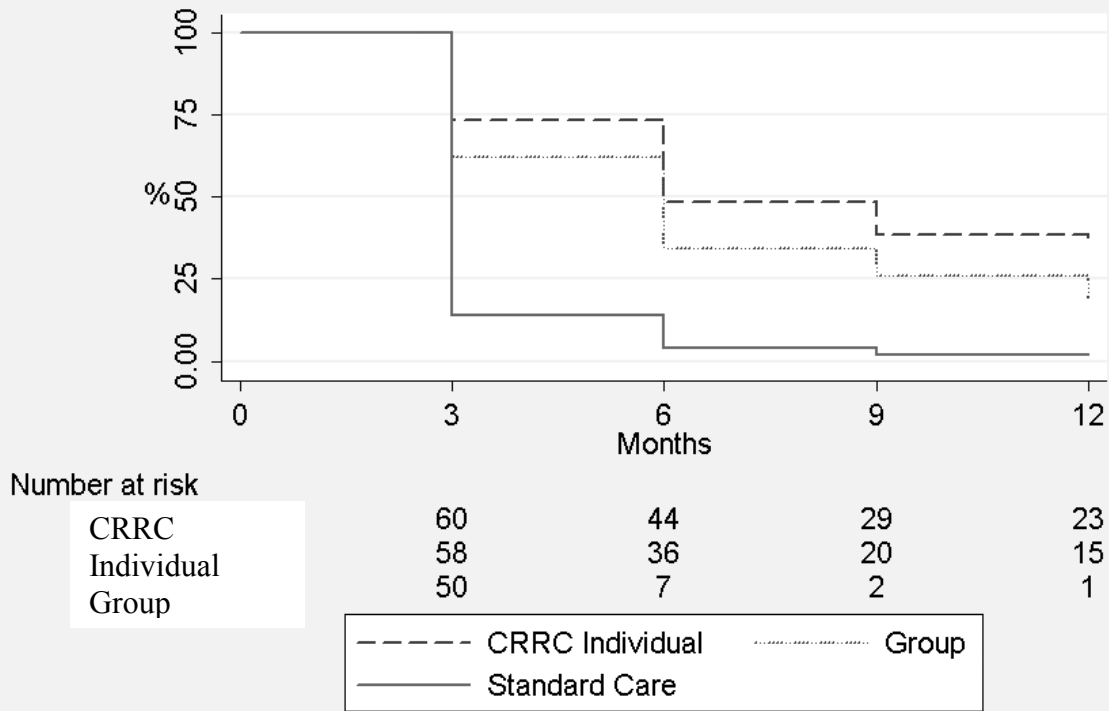


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1 Figure 2.

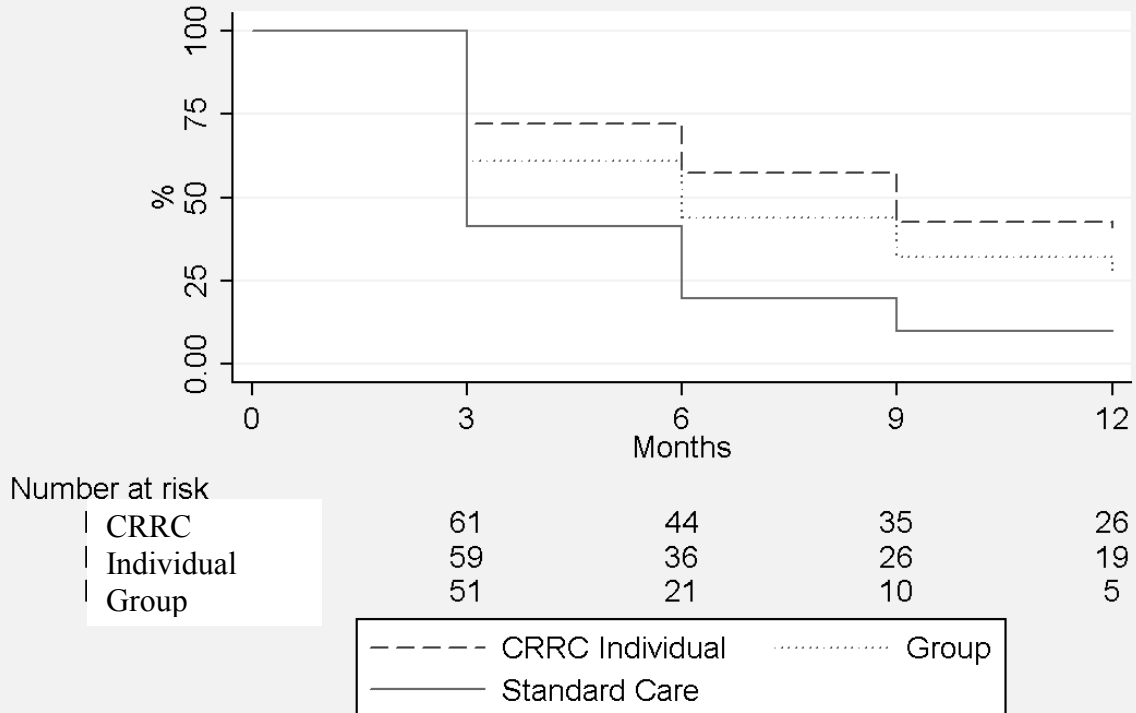
Accepted Manuscript

Kaplan-Meier survival estimates HBA1c Guideline Adherence



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Kaplan-Meier survival estimates Blood Pressure Guideline Adherence



2